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# Ethnobotanical studies of medicinal plants used by indigenous group of Hile region on Dhankuta District, Eastern Nepal

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**Abstract-** In Hile region of Dhankuta district, Nepal the ethnic people of villages belonging to the ethnic group- Chhetri, Rai, Limbu, Magar and Tamang, mainly depends on medicinal plants for their healthcare. In the present study, 35 informants from Limbu, Rai and Tamang ethnic group were selected to acknowledge the medicinal properties, preparation of drug, part of the plant used for drug preparation and use of medicinal plants for particular healthcare. Altogether, 26 plants belonging to 19 families were identified. The FIC value and FL (%) value were calculated. FIC value was calculated for 23 disease categories. The maximum value of FIC was 0.87 for Blood pressure and Liver disorder and the minimum FIC value was 0.29 for Dermatological problem. The FL (%) value was maximum for *Coelogyne corymbosa* (82.35) and *Eclipta prostata* (82.35) against Dermatological problem and Liver disorder respectively.

**Key words: Fidelity level (FL), Informants consensus factor (FIC), Ethnic group.**

## INTRODUCTION

Ethnobotany may be defined as scientific relationship between people and plants. Thus, ethnobotany include use of plants as food, shelter, rituals, clothing, medicine, etc. Utilization of plants for medicine purposes in India and Nepal has been documented long back in ancient literature.

Medicinal value of plants is mentioned in Rigveda and Atharvaveda. Several ancient Rishi described medicinal importance and use of plants for healing human beings. Charak is one of them. He wrote the book entitled "Charak Samhita" in which methods of preparation of drug from different part of plants has been described.

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Right from its beginning, the documentation of traditional knowledge on medicinal use of plants has provided many important drug of modern time.

The knowledge of drug formulation from plants acquired by indigenous people through experience was passed verbally from generation to generation. If the knowledge of drug formulation from plant known to indigenous people is documented, it will provide cost effective treatment without any side effect.

## METHODOLOGY

An ethnobotanical survey was conducted in different area of Hile, Dhankuta, Eastern Nepal. Hile is 13km north to Dhankuta at an elevation of 1948m. The major ethnic group of this locality are Chhetri, Rai, Limbu, Magar and

Tamang. Local people were consulted out of which 35 persons agreed to give proper information regarding medicinal plants of this area and their use in treatment of different diseases. Informants were both male and female belonging to the ethnic group Limbu, Rai and Tamang. The details of the informants is mentioned in Table 01.

**Table 1- Age & Gender distribution of Ethnic informants**

| Ethnic Group | Gender | | No. of Persons | Percentage (%) |
|---|---|---|---|---|
| | Male | Female | | |
| Limbu | 10 | 5 | 15 | 42.85 |
| Rai | 7 | 6 | 13 | 37.15 |
| Tamang | 4 | 3 | 7 | 20 |
| **Total** | 21 | 14 | 35 | |

Plant specimens were collected and their local name, season of the flowering of plant, part of the plant used for medicine and the disease for which it was used, acknowledged from informants.

The specimens were identified with the help of standard monograph.[1] Local names and medicinal uses were documented critically. The herbarium specimens were made accordingly.[2]

**Informants consensus factor (FIC):**

Informants consensus factor was calculated to find out the homogeneity in the information given by the informants. The *FIC* is calculated by the following formula.[3-6]

$$FIC = \frac{N_{ur} - N_t}{N_{ur} - 1}$$

Where, $N_{ur}$ is the number of use report in a particular illness category by informants and $N_t$ is the number of taxa or species used to treat that particular category by informants.

**Fidelity level (FL) value**

The fidelity level (FL), the percentage of informants claiming the use of a certain plants for the same major purpose was calculated according to the following formula.[7]

$$FL(\%) = \frac{I_P}{I_U} X\ 100$$

where $I_p$ is the number of informants who independently suggested the use of a plant species for a particular disease and $I_u$ is the total number of informants who mentioned the same plant for any disease.

## RESULT

An ethnobotanical survey was conducted in Hile region of Dhankuta district, Eastern Nepal. Local people were consulted. 35 informants were agreed to participate in present study. Informants belong to three ethnic groups- Limbu, Rai and Tamang. Both male and female informants were consulted at a regular visit in different seasons. The number of male informants were 10 from Limbu, 7 Rai, 4 Tamang and the female informants from ethnic group Limbu-5, Rai-6 and Tamang was 3. All informants were in between the age group of 30 to 65. The specimens were collected from field and herbarium were prepared. Each specimen was identified in the laboratory with the help of standard monograph. Local name, Medicinal value, parts of the plant for formulation of drug and the ailment were acknowledged from the informants and recorded. Altogether, 26 plants belonging to 19 families were identified. The maximum plants were from family Compositae (4). Name of the family and the no. of plants were recorded in table 4. *FIC* value and Fidelity level value were also determined which are recorded in table 02 and 03 respectively. *FIC* value was calculated for 23 disease categories. The maximum value of *FIC* was 0.87 for Blood pressure and Liver disorder and the minimum *FIC* value was 0.29 for Dermatological problem. The FL (%) value was maximum for *Coelogyne corymbosa* (82.35) and *Eclipta prostata* (82.35) against Dermatological problem and Liver disorder respectively.

**Table 2- Informant consensus factor (*FIC*) by categories of diseases**

| Disease category | Use Report ($N_{ur}$) | No. of Texa ($N_t$) | $F_{IC}$ |
|---|---|---|---|
| Asthma | 5 | 3 | 0.5 |
| Bleeding | 12 | 7 | 0.45 |
| Blood pressure | 9 | 2 | 0.87 |
| Bronchitis | 7 | 4 | 0.50 |
| Cardio-vascular disease | 6 | 2 | 0.8 |
| Cold and Cough | 22 | 12 | 0.47 |
| Dermatological diseases | 18 | 13 | 0.29 |
| Diarrhoea and Dysentery | 24 | 17 | 0.30 |
| Diuretic | 15 | 8 | 0.5 |
| Dyspepsia | 8 | 3 | 0.71 |
| Gastro-intestinal | 14 | 6 | 0.61 |
| Hormonal disorder | 7 | 2 | 0.83 |
| Leucorrhoea | 16 | 5 | 0.73 |
| Liver disorder | 9 | 2 | 0.87 |
| Lymphatic system | 4 | 2 | 0.67 |
| Musculoskeleton | 7 | 4 | 0.5 |
| Nervous system | 13 | 8 | 0.41 |
| Oral, Dental and ENT | 21 | 12 | 0.45 |
| Piles | 17 | 11 | 0.37 |
| Pulmonary | 5 | 2 | 0.75 |
| Skin disease and Intestinal disorder | 18 | 11 | 0.41 |
| Urinary disease | 22 | 15 | 0.33 |
| Venereal | 8 | 2 | 0.85 |

**Table 3- Fidelity level (FL) value of medicinal plants against a given Therapeutic category.**

| Medicinal Plant | Therapeutic Categories | $I_P$ | $I_U$ | FL% |
|---|---|---|---|---|
| *Bergenia ciliata* | Urinogenital and venereal | 12 | 16 | 75 |
| *Potentilla polyphylla* | Bleeding | 14 | 19 | 73.68 |
| *Centella asiatica* | Nervous system | 8 | 10 | 80 |
| *Camellia kissi* | Lymphatic system | 9 | 12 | 75 |
| *Cuscuta reflexa* | Oral, Dental and ENT | 14 | 20 | 70 |
| *Tinospora sinensis* | Gastro-intestinal | 16 | 21 | 76.19 |
| *Dendrodium moschatum* | Oral, Dental and ENT | 13 | 19 | 68.42 |
| *Vitex negundo* | Pulmonary | 8 | 10 | 80 |
| *Swertia chirayita* | Blood pressure | 15 | 20 | 75 |
| *Indigofera bracteata* | Hormonal disorder | 11 | 14 | 78.57 |
| *Sambucus adnata* | Cold and Cough | 16 | 22 | 72.72 |
| *Nephrolepis auriculata* | Cardio-vascular disease | 9 | 11 | 81.81 |
| *Coelogyne corymbosa* | Dermatological | 14 | 17 | 82.35 |
| *Zanthoxylum acanthopodium* | Musculoskeleton | 12 | 16 | 75 |
| *Acalypha indica* | Bronchitis | 15 | 19 | 78.94 |
| *Cicca disticha* | Liver tonic | 16 | 20 | 80 |
| *Phyllanthus niruri* | Urinogenital | 11 | 14 | 78.57 |
| *Evolvulus alsinoides* | Bronchitis and Asthama | 12 | 18 | 66.67 |
| *Eclipta prostata* | Liver tonic | 14 | 17 | 82.35 |
| *Ageratum conyzoides* | Diarrhoea and Dysentery | 14 | 20 | 70 |
| *Echinops hiatus* | Dyspepsia | 8 | 10 | 80 |
| *Vernonia cinerea* | Leucorrhoea | 10 | 13 | 76.92 |
| *Achyranthes aspera* | Urinary disease | 11 | 15 | 73.34 |
| *Rungia repens* | Diuretic | 9 | 12 | 75 |
| *Oxalis corniculata* | Piles | 8 | 11 | 72.72 |
| *Aloe vera* | Skin disease and Intestinal disorder | 16 | 22 | 72.72 |

**Table 4- Taxonomic diversity of medicinal plants in Hile region of Dhankuta, Eastern Nepal**

| Name of Family | No. of genera | No. of species | Percentage of species |
|---|---|---|---|
| Saxifragaceae | 1 | 1 | 3.84 |
| Rosaceae | 1 | 1 | 3.84 |
| Apiaceae | 1 | 1 | 3.84 |
| Theaceae | 1 | 1 | 3.84 |
| Convulvulaceae | 2 | 2 | 7.69 |
| Menispermaceae | 1 | 1 | 3.84 |
| Orchidaceae | 2 | 2 | 7.69 |
| Verbenaceae | 1 | 1 | 3.84 |
| Gentianaceae | 1 | 1 | 3.84 |
| Fabaceae | 1 | 1 | 3.84 |
| Adoxaceae | 1 | 1 | 3.84 |
| Davalliaceae | 1 | 1 | 3.84 |
| Rutaceae | 1 | 1 | 3.84 |
| Euphorbiaceae | 3 | 3 | 11.53 |
| Compositae | 4 | 4 | 15.38 |
| Amarantaceae | 1 | 1 | 3.84 |
| Acanthaceae | 1 | 1 | 3.84 |
| Oxalidaceae | 1 | 1 | 3.84 |
| Liliaceae | 1 | 1 | 3.84 |

**Table 5- Medicinal plant of Hile region, Dhankuata district, Eastern Nepal**

| Name of Plant | Local Name | Family | Medicinal use |
|---|---|---|---|
| *Bergenia ciliata* | Ghungri | Saxifragaceae | Urinogenital and venereal |
| *Potentilla polyphylla* | Bajradanti | Rosaceae | Bleeding |
| *Centella asiatica* | Ghortapare | Apiaceae | Nervous system |
| *Camellia kissi* | Banchiya | Theaceae | Lymphatic system |
| *Cuscuta reflexa* | Akash bel | Convulvulaceae | Oral, Dental and ENT |
| *Tinospora sinensis* | Gurgo | Menispermaceae | Gastro-intestinal |
| *Dendrodium moschatum* | Sugandha | Orchidaceae | Oral, Dental and ENT |
| *Vitex negundo* | Simali | Verbenaceae | Pulmonary |
| *Swertia chirayita* | Chiraito | Gentianaceae | Blood pressure |
| *Indigofera bracteata* | Sakhino | Fabaceae | Hormonal disorder |
| *Sambucus adnata* | Moti phul | Adoxaceae | Cold and Cough |
| *Nephrolepis auriculata* | Panisaro | Davalliaceae | Cardio-vascular disease |
| *Coelogyne corymbosa* | Jibanti | Orchidaceae | Dermato-logical |
| *Zanthoxylum acanthopodium* | Annkhe Timur | Rutaceae | Musculo-skeleton |
| *Acalypha indica* | Kuppee | Euphorbiaceae | Bronchitis |
| *Cicca disticha* | Narphal | Euphorbiaceae | Liver tonic |
| *Phyllanthus niruri* | Jaramla | Euphorbiaceae | Urinogenital |
| *Evolvulus alsinoides* | Shankhavati | Convolvulaceae | Bronchitis and Asthama |
| *Eclipta prostata* | Bhagrai | Compositae | Liver tonic |
| *Ageratum conyzoides* | Osari | Compositae | Diarrhoea and Dysentery |
| *Echinops hiatus* | Gokru | Compositae | Dyspepsia |
| *Vernonia cinerea* | Sadodi | Compositae | Leucorrhoea |
| *Achyranthes aspera* | Latjira | Amarantaceae | Urinary disease |
| *Rungia repens* | Parpatha | Acanthaceae | Diuretic |
| *Oxalis corniculata* | Khattibunti | Oxalidaceae | Piles |
| *Aloe vera* | Ghritkumari | Liliaceae | Skin disease and Intestinal disorder |

## CONCLUSION

The present study reveals that the three ethnic communities Limbu, Rai and Tamang depends on a variety of plants for their requirements to cure various diseases. They have a good knowledge about medicinal preparation, part of the plant from which medicine is prepared, mode of administration and medicinal doses. The Hile region consist of varieties of medicinal plants to cure several diseases. The local people use these plants against different diseases. In present investigation, altogether 26 medicinal plants were collected belonging to the 19 families. The chemical composition of these medicinal plants should be identified scientifically which will be great contribution for pharmaceutical and herbal industries.

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